

derived from the received waveform in the preceding bit interval by use of a 1-bit delay. In essence, the received waveform delayed by 1-bit duration serves as its own reference.

Data transmission using packet switching

This is done by supplying various addressed packets, which are interconnected to have the conversation. New dedicated paths are created for sending the data. From the multiple paths to the destination, any path can be used to send data. Cellular digital packet data was designed for optimal operation with an analogue cellular system, especially AMPS.

Short message service. Short message service is the most common packet service that is supported on digital cellular networks like GSM, IS-136, EDGE and PDC (packet data service). It is a store-and-forward/packet mode service that provides inter-working with the various applications and services within a fixed network. For message transfer between relevant network entities, control and signaling channels (instead of normal traffic channels) are generally used for data transmission.

General packet radio service (GPRS). GPRS essentially represents add-on capabilities to the basic voice-optimised cellular network that nevertheless maintain the essential characteristics of radio-access technology.

Enhanced data rates for GSM evolution (EDGE). In order to enhance the data handling capabilities of 2G service, radio-access portion had to be modified. This modification was evolved in Europe in the form of EDGE. EDGE also supports a link adaptation mechanism that selects the best combination of modulation and encoding schemes based on the time-varying link quality.

EDGE concept applies to both circuit-mode and packet-mode data and is sufficiently generic for application to other digital cellular systems. It works in the 200kHz bandwidth with one or more high-level modulation schemes and a range of efficient cod-

ing methods. Modulation schemes are offset QPSK and offset 16 QAM.

Spread spectrum

It is a special communication technique that purposefully uses much more RF bandwidth than necessary to transmit a signal. This helps in improving the signal-to-noise (S/N) ratio. The main advantages of this technique are secure communication and resistance to intentional jamming. There are 75 channels in the 2400-2483.3MHz band.

There are two methods of performing spread spectrum:

Frequency hopping. This technique spreads the narrow-band signal as a function of time. The transmitted frequency is changed to a different pre-assigned channel several times per second (hopped). The order in which the pre-assigned channels are selected is 'pseudo random.' In other words, the channel order is seemingly random but actually repeats itself at a defined interval. The specific order in which frequencies are occupied is a function of code sequence and the rate of hopping from one frequency to another is a function of information rate.

Direct sequence. This technique spreads a signal by expanding the signal over a broadband portion of the radio band. It uses a locally generated pseudo noise (PN) code to encode digital data to be transmitted. The most practical all-digit version is direct sequence. Binary phase-shift keying is the simplest and most often used modulation technique.

One of the most important features of spread-spectrum signals is that these contain a large number of very different signaling formats, used for communicating data symbols. It means that the receiver which detects one of these formats cannot detect any other format within a single message. The number of formats used in a spread-spectrum system is called multiplicity factor of the communication link and amounts to thousands.

CDMA. CDMA is a form of direct-sequence spread-spectrum technology that allows many users to occupy the

same time and frequency allocations in a given band/space. CDMA assigns each user a unique spreading code to spread the baseband data before transmission, in order to help differentiate signals from various users in the same spectrum. It is the platform on which 2G and advanced 3G services are built.

After speech, the codec converts voice into digital signal. CDMA spreads the voice stream over the full 1.25MHz bandwidth of the CDMA channel, coding each stream separately. The receiver uses a correlator to despread the wanted signal, which is passed through a bandpass filter. Unwanted signals are not despread and not passed through the filter.

The rate of the spreading signal is known as the 'chip rate' as each bit in the spreading signal is known as 'chip.' All 2G networks support only single-user data rates of the order of 10 kbps, which is too slow for rapid e-mail and Internet browsing.

CDMA provides more than ten times the capacity of the analogue AMPS and five times the calling capacity of GSM and TDMA systems. It requires fewer cell sites than GSM and TDMA.

Personal communication system

Personal communication system (PCS) is a new class of cellular telephone system such as AMPS. PCS systems are a combination of cellular telephone network and intelligent network, which is the entity of super-simple transfer (SST) inter-office protocol that distinguishes physical components of the switching network such as signal service point, signal control point and signal transfer point from the services provided by SST network.

In essence, PCS is the North American implementation of European GSM standard. GSM utilised its own TDMA access methods and provided expanded capacity and unique services such as caller ID, call forwarding and short messaging. A critical feature was seamless roaming, which allowed subscribers to move across provider boundaries.